

Technical Datasheet VBF-T2R1SINGLEMODEL-DSH-01

Virtual Battery Factory — t2_r1_singlemodel — 2026-08-25

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> Case t2_r1_singlemodel — grid energy storage cell (virtual design). Generation date 2026-08-25.

Field	Value	Source
Rated capacity	5.0 Ah (parameter set nominal); 6.9555 Ah (simulated 1C discharge)	parameter set / cell/r3_1c_dfn.json
Nominal voltage / window	2.5 – 4.2 V; midpoint 3.574 V	parameter set / cell/r3_energy.json
Rated energy	25.03 Wh	cell/r3_energy.json (V*I integration)
Energy density	358.0 Wh/kg (contract caliber, electrolyte excluded)	cell/r3_energy.json
Max continuous discharge rate	1C verified (6.9555 Ah); 5C 0.4939 Ah (7% of 1C; voltage collapses to cut-off in ~71 s — severely rate-limited; informational, no 5C criterion)	cell/r3_1c_dfn.json, cell/r3_5c_dfn.json
Fast-charge capability	4C CC to 4.2 V: 5.57 Ah charged, anode potential min +0.112 V (no plating), T_max 342.5 K	cell/r3_4c45.json
Operating temperature range	-20 C discharge verified (retention 99.6 %); 45 C charge verified; storage limits Not provided	cell/r3_lowT.json / cell/r3_4c45.json
Cycle life (SEI)	194.6 nm @100 cyc; 439.7 nm @500 cyc (DFN aging, ec reaction limited; capacity trajectory shows the standard climb artifact — SEI thickness is the reliable indicator)	cell/r3_aging100.json / cell/r3_aging500_d1e19.json
Safety determination	4C plating-free (virtual); overcharge-coupled TR not triggered (partial-run T_max 322.7 K; DFN overcharge 4.7 V charge phase unsolvable — recorded verbatim); 5 W nail triggers TR (T_max 408.4 K)	cell/r3_tr.json / cell/r3_tr_nail.json
Dimensions and mass	65 x 1580 x 0.390 mm (layer stack); 69.9 g (electrolyte excluded)	parameter set / cell/r3_energy.json